# Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Fiscal Policy — qualifier level (Listas 5–6)

**Instructions.** Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

## Block 1 — General equilibrium, hours and taxes

*(Kurlat ch. 9; Romer ch. 2 • 25 points)*

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**Question 1.** (10 points) Objective items.

**a)** (3 points) *General equilibrium × labor supply.* A two-period economy has a fixed capital stock $K$ and no investment. Households have $u(c) + \psi \ln l$ each period, $u = c^{1-\sigma}/(1-\sigma)$, one unit of time; firms produce $Y = AK^{\alpha}L^{1-\alpha}$ competitively. In equilibrium, with $L = 1/3$, $\sigma = 2$ and $\alpha = 1/3$, the elasticity $d\ln L/d\ln A$ is:

A. $-0.46$

B. $0$

C. $+0.46$

D. $-1.00$

**b)** (3 points) *General equilibrium × growth.* In the continuous-time Ramsey model with $f(\tilde{k}) = \tilde{k}^{1/3}$, $\rho = 0.03$, $\sigma = 2$, $g = 0.02$, $n = 0.01$, $\delta = 0.05$, the steady-state gross saving rate is:

A. 33.3%

B. 22.2%

C. 26.7%

D. 16.7%

**c)** (2 points, T or F) *General equilibrium × fiscal policy.* In the Ramsey model, a permanent, unanticipated rise in government purchases financed by lump-sum taxes lowers consumption one for one on impact and leaves capital unchanged forever.

**d)** (2 points, T or F) *General equilibrium × taxes.* A permanent tax on capital income whose revenue is rebated lump-sum leaves the steady-state capital stock unchanged, because the rebate gives the revenue back.

**Question 2.** (15 points) Discursive. *General equilibrium × labor supply × taxes.* An infinite-horizon economy without growth: households maximize $\int_0^{\infty} e^{-\rho t}[\ln c + \psi \ln(1-h)]\,dt$, own the capital and rent it to firms; firms produce $Y = K^{\alpha}h^{1-\alpha}$; capital depreciates at $\delta$; labor income is taxed at $\tau$ and the revenue is rebated lump-sum. Numbers: $\alpha = 1/3$, $\rho = 0.04$, $\delta = 0.06$; $\psi$ is set so that $h = 1/3$ when $\tau = 0.3$.

**a)** (3 points) Derive the household's optimality conditions (intertemporal and intratemporal), the firm's conditions and the government budget, and write the equilibrium as a dynamic system in $(K, c)$.

*Solution:*

**b)** (3 points) Find the steady state. Which ratios does the tax leave unchanged? Find hours in closed form and calibrate $\psi$.

*Solution:*

**c)** (3 points) The tax rises permanently and unexpectedly from 0.30 to 0.45. Find the long-run changes in $h$, $K$, $Y$, $C$, $r$ and $w$.

*Solution:*

**d)** (3 points) Characterize the transition. Linearize the system at the new steady state (use the implicit function for $h$), show that it is a saddle, find the half-life, the impact responses of $c$ and $h$, and draw the paths of $K$, $c$ and $h$ (percent deviations from the old steady state).

*Solution:*

**e)** (3 points) *Parameter analysis.* How do steady-state hours and the speed of convergence depend on $\tau$ and $\psi$? Prove the signs you can, fill the table for $\tau \in \{0, 0.3, 0.45, 0.6\}$, give the limits,

and explain why the speed changes.

*Solution:*

## Block 2 — Money, inflation and fiscal policy

*(Kurlat chs. 10–11; Romer ch. 11 • 25 points)*

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**Question 3.** (10 points) Objective items.

**a)** (3 points) *Money demand × welfare.* Baumol–Tobin: income $Y = 1200$ a year, cost $F = 2$ per trip to the bank, nominal rate $i = 6\%$. The welfare cost of $i > 0$, measured as the Bailey area $\int_0^i m(x)\,dx - i\,m(i)$, is:

A. 8.49

B. 16.97

C. 4.24

D. 0

**b)** (3 points) *Money × expectations.* Cagan with perfect foresight, in logs: $m_t - p_t = -a(p_{t+1} - p_t)$, $a = 4$ (monthly). At $t = 0$ the central bank credibly announces a permanent 10% rise in the money supply at $t = 4$. On impact, $p_0$ rises by:

A. 0%

B. 4.1%

C. 10%

D. 8.0%

**c)** (2 points, T or F) In the Cagan model with perfect foresight, a credible announcement that money growth will fall permanently in the future leaves inflation unchanged until money growth actually falls.

**d)** (2 points, T or F) *Money × taxes.* If nominal interest income is taxed at 30%, keeping the after-tax real return constant requires the nominal rate to rise by more than one point per point of expected inflation.

**Question 4.** (15 points) Discursive. *Money × fiscal policy × expectations: unpleasant arithmetic.* Real money demand (share of GDP) is $m = \kappa e^{-a\pi}$, $\kappa = 0.3$, $a = 2.5$ (annual); output does not grow, so seigniorage in a steady state is $S = \pi m$. Debt (share of GDP) evolves as $b_{t+1} = (1 + r)b_t + d - S_t$, with $r = 0.05$, primary deficit $d = 0.005$ and $b_0 = 0.3$. For $T$ years the central bank holds seigniorage at $S_1$; afterwards debt is frozen and seigniorage covers the interest bill.

**a)** (3 points) Characterize the steady states of this economy: the relation between debt, deficit and seigniorage, the maximum seigniorage, and the inflation rate if debt is stabilized at once.

*Solution:*

**b)** (3 points) Find debt at $T$ and the seigniorage needed afterwards in closed form. Compute them and the inflation rates during and after the tight phase for $S_1 = 0.01$, $T = 10$.

*Solution:*

**c)** (3 points) Prove that tighter money today means higher inflation later, and find the length of the tight phase beyond which no steady state exists.

*Solution:*

**d)** (3 points) Now money demand is forward-looking, $m_t - p_t = -a(p_{t+1} - p_t)$ in logs, money grows at $\mu_1$ until $T$ and at $\mu_2$ afterwards, and this is known. Find inflation in the years before $T$ and evaluate it 1, 3 and 5 years before the switch with $\mu_1 = 3.65\%$ and $\mu_2 = 11.76\%$.

*Solution:*

**e)** (3 points) *Parameter analysis.* How does post-stabilization inflation depend on $S_1$, $T$ and $r$? Prove the signs, fill the table for $S_1 \in \{0.005, 0.01, 0.015\}$ and $T \in \{5, 10, 20\}$, compute $T^*$ for each $S_1$, and draw $\pi_2$ against $T$ for the three $S_1$.

*Solution:*